

Total No. of Questions : 6]

SEAT No. :

P4909

[Total No. of Pages : 2

B.E./Insem. - 62
B.E. (IT)
MACHINE LEARNING
(2012 Pattern) (Semester - I)

Time :1Hour]

[Max. Marks :30

Instructions to the candidates:-

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1)** a) What is machine learning? Give an overview of machine learning with suitable diagram. [5]
- b) Explain geometric models and probabilistic models of machine learning in detail. [5]

OR

- Q2)** a) Explain in detail : [4]
- i) Supervised Learning vs Unsupervised learning
 - ii) Training dataset vs testing dataset
- b) Explain features with suitable example. Give two uses of features. Explain the term discretization of features. [6]
- Q3)** a) Define following terms with suitable example : [6]
- i) Confusion matrix
 - ii) False positive rate
 - iii) True positive rate
- b) What is overfitting? Specify the reasons for overfitting. [4]

OR

- Q4)** a) What is a contingency table/Confusion Matrix? What is the use of it?[4]
- b) Derive and explain output code matrix for One Vs One and One Vs Rest Scheme for construction of Multi class classifier (for 3 classes) [6]

P.T.O.

- Q5)** a) Explain regression using least square method. [6]
b) What are support vectors and margins? Also explain Soft margin SVM. [4]

OR

- Q6)** a) Explain Perceptron training algorithm for linear classification. [6]
b) What is multivariate regression? Explain its equation using homogeneous coordinates. [4]

